Polymap: Lockstep-Wallet Discovery and On-Chain Funding Validation on Polymarket using Temporal Graph Learning

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by

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CERTIFICATE

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ABSTRACT

Prediction markets that settle on public blockchains expose every trade and every wallet transfer, yet pseudonymous wallets make it difficult to tell whether tightly synchronised trading reflects shared control, shared information or coincidence. This project presents Polymap, an exploratory pipeline for the Polymarket prediction market that keeps three analyses deliberately separate. First, a model-free lockstep screen selects wallet pairs whose trades repeatedly co-occur within seconds at near-identical prices. Second, a funding-validation design compares these candidate pairs with anchor-fixed controls matched on co-timing intensity, using observed Polygon transfers and wallet-level bootstrap intervals. Third, a separate memory-based temporal model with causal pair-history features ranks future co-trading links against hard negatives on a chronological held-out period. The pipeline was applied to two 63-day geopolitical markets: Xi (23,839 trades, 397 candidate pairs) and Iran (35,135 trades, 1,512 candidate pairs). Candidate pairs showed higher direct-transfer rates than controls in both markets (+6.8 and +4.6 percentage points), but neither primary 95% interval excluded zero, so the pre-agreed success criterion was not met. Excluding wallets with truncated transfer histories produced gaps of +9.9 and +5.6 points whose intervals excluded zero only narrowly, while shared-funder evidence did not replicate. The tuned temporal model improved test mean reciprocal rank over its plain variant in both markets (0.586 versus 0.341 on Xi; 0.321 versus 0.156 on Iran), exceeded a recency baseline on Xi (0.433) in every seed, and was competitive with it on Iran (0.330). The results are therefore suggestive rather than confirmed. The report documents test-set reuse, confounded ablations and coverage limitations, and stresses that no result establishes common ownership, coordination or intent.

Keywords: Prediction Markets • Polymarket • On-Chain Analytics • Wallet Clustering • Temporal Graph Networks • Link Prediction • Matched Controls

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Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
AP	Average Precision
AUC	Area Under the ROC Curve
CI	Confidence Interval
CLI	Command-Line Interface
CSV	Comma-Separated Values
CTF	Conditional Tokens Framework
DEX	Decentralised Exchange
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
GNN	Graph Neural Network
GRU	Gated Recurrent Unit
Hits@k	Fraction of queries with the true target ranked in the top k
JODIE	Joint Dynamic User–Item Embeddings
JSON	JavaScript Object Notation
MLP	Multilayer Perceptron
MRR	Mean Reciprocal Rank
pp	Percentage points
ReLU	Rectified Linear Unit
SD	Standard Deviation
SF	Shared Funder
ST-GNN	Spatio-Temporal Graph Neural Network
TGAT	Temporal Graph Attention Network
TGB	Temporal Graph Benchmark
TGN	Temporal Graph Network
USDC	USD Coin (stablecoin)
UTC	Coordinated Universal Time
YAML	YAML Ain’t Markup Language

Chapter 1
Introduction

1.1 Background of the project topic

Prediction markets allow participants to buy and sell contracts whose payoff depends on the outcome of a future event. Because a contract on a binary outcome pays one unit if the event occurs and nothing otherwise, its price is commonly interpreted as the market's probability estimate for that event [1]. Polymarket is currently the largest on-chain prediction market. Its contracts are conditional tokens recorded on the Polygon blockchain, orders are matched through a hybrid off-chain order book, and positions are settled in the USDC stablecoin [2, 3]. As a result, every executed trade and every token transfer between wallets is publicly visible.

This transparency creates an unusual research opportunity. In conventional financial markets, the identity of traders is hidden from researchers and visible only to regulators. On Polymarket, trades are attached to pseudonymous wallet addresses that anyone can inspect, and the funding history of each wallet can be traced on chain. At the same time, pseudonymity means that one person or organisation may control many wallets, and that the relationship between wallets is never stated explicitly. Understanding whether apparently independent wallets behave as if they are related is therefore both feasible and important for market integrity.

1.2 Motivation and scope of the report

Recent studies have shown that a substantial fraction of Polymarket volume may be artificial. Sirolly *et al.* used an iterative network method to identify clusters of wallets that repeatedly trade against each other and attributed roughly a quarter of historical volume to activity consistent with wash trading [4]. Dubach analysed a tick-level order-book archive joined to on-chain fills and reported a lower-bound self-counterparty wash share with a median of about 1% per market [3]. Both approaches focus on wallets that trade *with each other*.

This project studies a different behavioural pattern which we call **lockstep trading**: two distinct wallets that repeatedly trade the same market within seconds of each other and at nearly identical prices. Lockstep wallets need not be counterparties, so counterparty-based detectors do not see them. Lockstep behaviour can arise from one operator splitting orders across several wallets, from coordinated groups similar to those observed in cryptocurrency pump-and-dump campaigns [5],

from automated bots reacting to the same public signal, or simply by chance in a busy market. Because the explanations differ so much, trading similarity alone cannot establish any relationship between wallets.

On-chain funding provides an independent source of evidence. If two lockstep wallets have transferred money directly to each other, or have been funded by the same uncommon source, this is consistent with (though not proof of) a relationship. Funding links of this kind are the foundation of classical blockchain address-clustering heuristics [6, 7]. However, exchanges, bridges and routing contracts fund very large numbers of unrelated wallets, so funding links are only meaningful when they are compared with an appropriate reference group.

In parallel, graph neural networks have been applied to cryptocurrency fraud [8, 9], and memory-based temporal graph networks have become a standard tool for modelling evolving interaction graphs [10, 11]. Benchmarks have also shown that very simple memorisation heuristics can be surprisingly competitive with these models on cryptocurrency data [12, 13]. The scope of this report is therefore twofold: (i) to test, with a model-free screen and matched controls, whether lockstep wallet pairs show more observable Polygon funding links than comparable pairs; and (ii) to evaluate, as a separate experiment, whether a memory-based temporal model can rank future co-trading links better than a strong recency baseline. The study covers two geopolitical Polymarket markets, referred to as *Xi* and *Iran*, each observed for 63 days.

1.3 Problem statement

Given the public trade stream of a Polymarket market and the Polygon transfer histories of the wallets involved, the project addresses the following question: **do wallet pairs that are screened for close co-trading show observable Polygon funding links more often than comparable wallet pairs?** The candidate screen must not depend on any trained model, so that a model is never used to select the pairs that are later used to validate it. A secondary question is whether a custom memory-based temporal model with causal pair-history features can rank future co-trading links better than non-learned baselines on a chronological held-out period. Throughout, the permitted interpretation is limited to “lockstep pair” and “funding link observed”; the project does not claim to identify common ownership, coordination or intent.

1.4 Salient contribution

The main contributions of this project are:

- A **model-free lockstep screen** that selects candidate wallet pairs from temporal co-trading edges using explicit time, price and persistence thresholds, together with stratified permutation checks.
- An **anchor-fixed, co-timing-matched control design** for on-chain funding validation, with a primary measure (direct transfer) that does not depend on any hub threshold, and wallet-level bootstrap intervals that respect the dependence between pairs that share wallets.
- A **memory-based temporal link-ranking model** with six causal pair-history features, evaluated online against exact shared hard negatives and a recency baseline over five seeds in two markets.
- A **two-market replication** judged against a pre-agreed success criterion and reported in full, including a negative primary result, truncation sensitivity analyses, test-set reuse and confounded ablations.
- A modular Python pipeline with a command-line interface, saved artifacts and an existing React wallet-graph explorer, with a documented plan for a funding-evidence visualisation.

1.5 Organization of report

The remainder of this report is organised as follows. Chapter 2 reviews related work on prediction-market integrity, wash trading and address clustering, graph learning for cryptocurrency fraud, temporal graph neural networks and their evaluation, and the statistical methods used. Chapter 3 describes the system architecture, computing environment, candidate screen, funding-validation design, temporal model, datasets, implementation, evaluation metrics and the overall algorithm. Chapter 4 reports and analyses the results of the permutation checks, funding validation and link-ranking experiments, and discusses error patterns. Chapter 5 discusses advantages, limitations and applications. Chapter 6 concludes the report and outlines future work. References and appendices containing the algorithm, specifications, source files, raw observations and hardware notes follow.

1.6 Summary

This chapter introduced Polymarket as a transparent but pseudonymous prediction market, motivated the study of lockstep co-trading wallets and their on-chain funding links, stated the research problem, summarised the principal contributions, and explained the organisation of the report.

Chapter 2
Literature Survey

2.1 Introduction

The purpose of this literature survey is to understand the progress, techniques and open challenges in the areas that this project draws on. The survey is organised into five streams: prediction-market integrity, wash trading and address clustering on blockchains, graph learning for cryptocurrency fraud, temporal graph neural networks and their evaluation, and statistical design for observational validation. For each stream we summarise representative studies and identify the gap that motivates this project.

2.2 Prediction Markets and Market Integrity

Wolfers and Zitzewitz [1] reviewed the economics of prediction markets and showed that market prices on contracts tied to elections, economic indicators and other events often forecast outcomes at least as well as polls or expert opinion. They also discussed how prices can be interpreted as probabilities and what conditions are required for this interpretation to hold. A natural concern is manipulation: if a trader with a strong preference moves prices away from the consensus, the informational value of the market may decline. Hanson and Oprea [14] analysed this concern theoretically and showed that, under some conditions, a manipulator can actually *improve* accuracy because the manipulation creates profit opportunities that attract informed traders. Manipulation is therefore not automatically harmful to price accuracy, but it does distort volume and the appearance of market interest.

Polymarket differs from earlier prediction markets because it is built on a public blockchain. Each binary market is represented by two conditional tokens (YES and NO), trading occurs through an exchange contract that settles signed orders on chain, and USDC is the medium of exchange [2]. In 2026, Dubach [3] studied the microstructure of Polymarket using a continuous archive of the public order-book feed covering 52 days and more than 30 billion events, joined to 255 million on-chain fill events, on a pre-registered panel of 600 markets. The study reported eight stylised facts, including a longshot spread premium and broad maker-wallet diversity. Two findings are directly relevant here. First, trade direction inferred from the public feed agreed with on-chain ground truth only about 59% of the time, so on-chain records must be used as the authoritative

source of trades. Second, a strict self-counterparty wash detector found a median wash share of 0.97% per market with a maximum of 22.2%, explicitly described as a lower bound because multi-wallet patterns were not detected.

Sirolly, Ma, Kanoria and Sethi [4] proposed an iterative network-based procedure for unsupervised wash-trade detection on Polymarket. Their key idea is that wash traders form approximately closed clusters of counterparties that rarely trade with other participants. They first identify wallets that repeatedly close their positions and then look for clusters of such wallets trading against each other. Applied to Polymarket's history, the method attributed roughly 25% of total volume to activity consistent with wash trading, with a peak near 60% of volume in December 2024, and flagged about 14% of the 1.26 million wallets that had ever traded. These studies establish that wallet-level behaviour on Polymarket can be analysed at scale, but both focus on trades *between* related wallets. Same-side, near-simultaneous trading by distinct wallets that never face each other is not measured.

2.3 Wash Trading and Address Clustering on Blockchains

Wash trading has been studied extensively on cryptocurrency exchanges. Cong, Li, Tang and Yang [15] used statistical regularities that authentic trading is expected to satisfy, such as patterns in trade-size distributions and round-number clustering, to estimate wash trading on unregulated centralised exchanges, finding that a large share of reported volume on many such exchanges was likely fabricated. Victor and Weintraud [16] studied decentralised exchanges on Ethereum and detected wash trading by searching for sets of accounts whose trades form closed cycles with little or no net change in position. Both approaches identify wash trading through counterparty relationships within the trade data itself.

A separate line of work links pseudonymous addresses into clusters that are likely controlled by the same entity. Meiklejohn *et al.* [6] introduced heuristics for Bitcoin, most notably that all inputs to a transaction are controlled by the same user and that one-time change addresses belong to the sender, and used them to trace flows of funds between services. Ethereum uses an account model rather than transaction inputs, so these heuristics do not apply directly. Victor [7] proposed Ethereum-specific heuristics based on deposit-address reuse at exchanges, participation of several addresses in the same airdrop, and token-approval relationships, and showed that they can link a meaningful fraction of active addresses. These heuristics treat funding and transfer relationships

as evidence of common control. Their main weakness is that infrastructure addresses (exchanges, bridges, routers and custodians) interact with very large numbers of unrelated users, so naive common-funder rules produce many false links.

Coordinated behaviour is also documented outside wash trading. Xu and Livshits [5] studied cryptocurrency pump-and-dump schemes organised through public messaging channels, characterised how organisers coordinate participants to buy a target coin at a set time, and built a model to predict likely targets. Their work shows that coordinated groups leave timing footprints in market data, which motivates the use of close co-timing as a screening signal.

2.4 Graph Learning for Cryptocurrency Fraud Detection

Akoglu, Tong and Koutra [17] surveyed graph-based anomaly detection and organised methods according to whether graphs are static or dynamic, attributed or plain, and whether labels are available. Graph structure is particularly useful in financial fraud because fraudulent behaviour is usually relational. Weber *et al.* [8] released the Elliptic dataset of more than 200,000 Bitcoin transactions, a small fraction labelled illicit and a larger fraction labelled licit, and compared graph convolutional networks with logistic regression and random forests for anti-money-laundering classification. They found that graph structure helped, but that random forests on node features remained strong, and highlighted difficulties when the data distribution shifted over time.

In 2026, Losavio, Persia, Sathe and Pasadakis [9] addressed pump-and-dump detection when only aggregated hourly market data (open, high, low, close, volume and trade counts) are available, so that relationships between tokens are not observed directly. They proposed three ways of inferring a token graph: a static correlation graph built from the training period, an event-driven dynamic correlation graph updated around known fraud events, and a self-adaptive adjacency matrix learned from node embeddings. Each graph was processed by the same spatio-temporal GNN that combines Transformer-style graph attention with a temporal Transformer encoder. On a dataset of 84 tokens and 1,905,850 hourly observations with only 314 positive pump hours (about 0.016%), the self-adaptive graph achieved an F1-score of 0.62, compared with 0.53 for random forests, 0.49 for XGBoost and 0.56 for the best temporal-only GRU baseline. The authors used a chronological 60/20/20 split with a 5-hour embargo and argued that evaluating on short windows around known events inflates performance. Their emphasis on chronological evaluation and realistic class imbalance directly informs this project. However, their setting relies on labelled events and

inferred token-level graphs, whereas Polymarket wallets carry no ground-truth labels and wallet-level trades are observed directly.

2.5 Temporal Graph Neural Networks and Their Evaluation

Kazemi *et al.* [18] surveyed representation learning for dynamic graphs, distinguishing discrete-time snapshot models from continuous-time models that process timestamped events. The co-trading graph in this project is naturally continuous-time: each edge is created by a pair of trades at a specific moment.

Kumar, Zhang and Leskovec proposed JODIE [19], which maintains embeddings for users and items that are updated by two coupled recurrent networks whenever an interaction occurs, and uses a projection operator to estimate how an embedding drifts between interactions. Xu *et al.* proposed TGAT [11], which computes node embeddings as functions of time using self-attention over a node's temporal neighbourhood. TGAT introduced a functional time encoding derived from Bochner's theorem so that time differences can be embedded in a way compatible with attention, and it supports inductive inference for nodes not seen during training. It was evaluated on Reddit, Wikipedia and a large industrial user–product dataset.

Rossi *et al.* [10] proposed Temporal Graph Networks (TGN), a general framework that combines a per-node **memory** with message, aggregation, memory-update and embedding modules. When an event involves a node, a message is computed and the node's memory is updated by a recurrent cell such as a GRU [20]. The embedding module aggregates information from neighbours to reduce *memory staleness*, which occurs when a node has been inactive for a long time. The authors showed that JODIE, TGAT and other models can be cast as special cases of TGN. On future edge prediction, TGN with graph attention reported transductive average precision of 98.46% on Wikipedia and 98.70% on Reddit, compared with 95.34% and 98.12% for TGAT.

These high scores depend strongly on evaluation design. Poursafaei *et al.* [12] showed that the common practice of pairing each positive edge with a single randomly sampled negative edge makes link prediction too easy. They introduced *historical* negatives (pairs that interacted before but not at the current time) and *inductive* negatives, and proposed **EdgeBank**, a non-learned baseline that simply memorises previously observed edges. EdgeBank was competitive with many learned models under random-negative evaluation. Huang *et al.* then released the Temporal Graph

Benchmark (TGB) [13], which ranks each positive edge against many negatives (including historical negatives) and reports mean reciprocal rank (MRR). On `tgbl-coin`, a graph of stablecoin ERC-20 transfers between addresses from April to November 2022, TGN achieved a test MRR of 0.586, close to the time-windowed EdgeBank baseline at 0.580 and well above DyRep at 0.452. TGB therefore recommends always reporting EdgeBank-style baselines. Classical link-prediction scores such as common neighbours [21] provide further simple references.

2.6 Statistical Design for Observational Validation

Because no ground-truth labels are available, this project validates lockstep pairs observationally. Matching methods [22] compare a treated group with controls chosen to be similar on relevant covariates, here co-timing intensity. Uncertainty is quantified by bootstrap resampling [23]; for dependent event streams a block bootstrap that resamples contiguous blocks is appropriate [24]. When an observed statistic is compared with a small number of random permutations, the empirical p-value should include an add-one correction so that it is never zero [25]. Fixing primary outcomes and success criteria in advance reduces the risk of selective reporting [26], multiple related tests require care in interpretation [27], and repeated use of a test set is a recognised form of leakage in machine-learning research [28].

Table 1. Literature streams and the gap addressed by this project

Stream	Representative work	Gap addressed by this project
Prediction-market integrity	[1], [4], [3]	Detectors target counterparties trading with each other; same-time co-trading by distinct wallets is not measured
Wash trading and address clustering	[15], [16], [6], [7]	Funding links are treated as evidence without matched controls; hub funders saturate
Graph learning for crypto fraud	[8], [9]	Requires labelled events; Polymarket wallet graphs are unlabelled
Temporal graph networks	[19], [11], [10]	Learned models may not beat memorisation heuristics
Temporal-graph evaluation	[12], [13]	Use exact shared hard negatives, online scoring and a recency reference

2.7 Summary

The literature shows that wallet-level behaviour on Polymarket can be analysed at scale, that funding links are a standard but noisy signal of common control, that graph learning is useful for

cryptocurrency fraud, and that temporal graph models must be compared with strong memorisation baselines under hard-negative protocols. No reviewed study examines same-side, near-simultaneous co-trading by distinct Polymarket wallets and validates it against matched controls using on-chain funding evidence, which is the gap this project addresses.

Chapter 3
Methodology and Experimental Setup

3.1 Block Diagram

The proposed system is organised as a pipeline with two analytical branches that share only the cached trade data, the temporal co-trading edges and the candidate wallet list (Fig. 1). The **funding-evidence branch** selects candidate pairs with a model-free screen, collects Polygon transfers for the candidate wallets, constructs matched controls and estimates funding-link rates with bootstrap intervals. The **link-ranking branch** trains a memory-based temporal model on the candidate-wallet graph and evaluates it on a held-out future period. The trained model never selects the funding candidates.

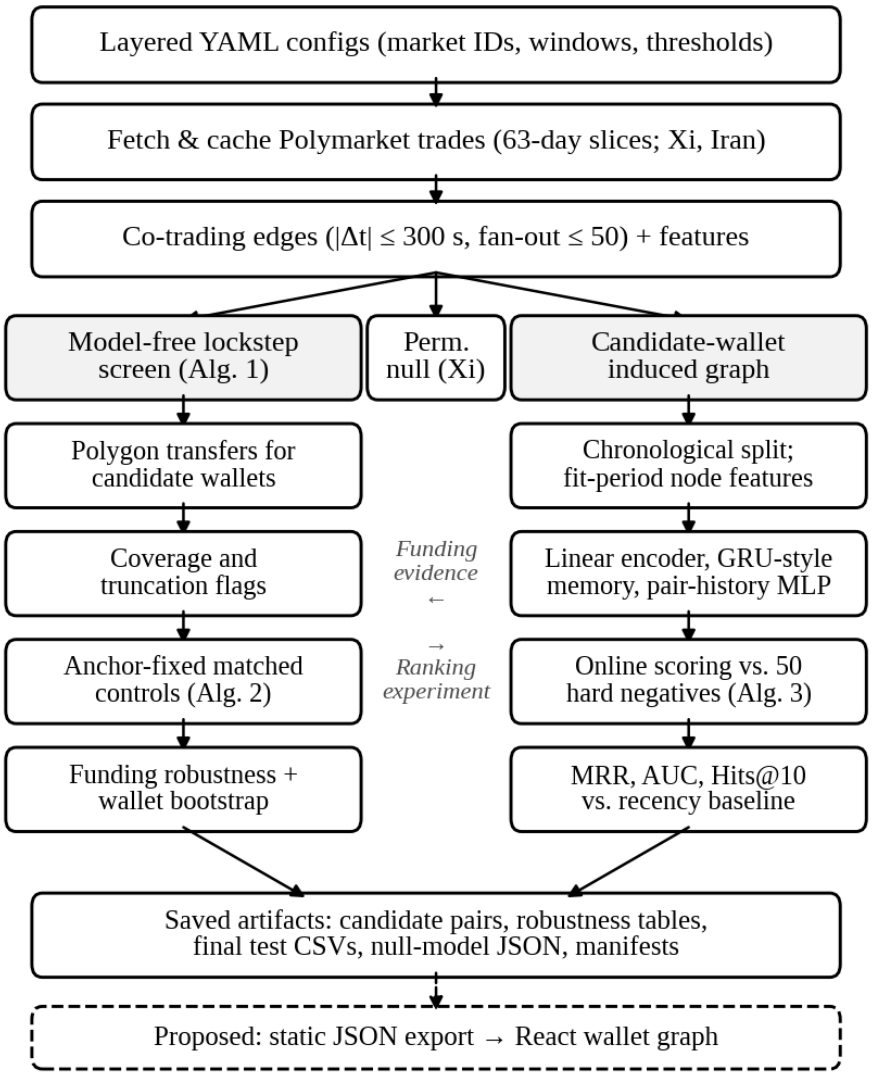


Figure 1. Block diagram of the Polymap system architecture

Because conclusions about real, identifiable wallet addresses are sensitive, a set of research invariants was enforced throughout the implementation (Table 2).

Table 2. Research invariants and how they are enforced

ID	Invariant	Enforcement
R1	Candidate selection never uses the trained model	`candidates` stage reads edge aggregates only
R2	Controls share an anchor wallet, match co-timing count, are non-candidates and are not reused	Anchor-fixed matcher (Algorithm 2)
R3	Missing coverage is not evidence of no link	Uncovered pairs dropped and counted; truncation flags
R4	Primary measure is independent of any hub threshold	Direct transfer at any time in window, Eq. (3)
R5	Node features use fit-period trades only	Train-period feature rebuild
R6	A test batch is scored before it updates model state	Online evaluator (Algorithm 3)

ID	Invariant	Enforcement
R7	Model and baselines see identical candidates	Shared batch boundaries and exact negative IDs
R8	No output claims ownership, coordination or intent	Report vocabulary and interface labels

3.2 Hardware Description

This is a software-only study. All final model runs were executed on CPU (`model.device = cpu`) on a local development workstation, and the analysis stages operate on cached files on disk. Trade data were retrieved from Polymarket's public trade interface and on-chain transfer histories from Polygon for the candidate wallets only. Table 3 summarises the computing environment. No custom sensing circuit, microcontroller or pin-level hardware is part of this project; consequently, a pin diagram is not applicable (Appendix V).

Table 3. Computing and software environment

Component	Description
Compute device	CPU only for final model runs (`model.device = cpu`)
Processor / memory	[To be filled in: workstation processor, RAM and operating system]
Programming language	Python (package `polymap`, installed in a project virtual environment)
Configuration	Layered YAML files (`configs/default.yaml` plus market overlays) with build fingerprints
Data formats	Parquet (trades, edges, candidates, transfers), CSV (results), JSON (manifests, null model)
Data sources	Polymarket public trade data; Polygon native and ERC-20 transfer histories
Interface	Command-line stages (`polymap <stage>`), backend API, React/Vite wallet-graph explorer
Testing	57 passing local tests; syntax and targeted lint checks

3.3 Software Description, Flowchart / Algorithm

3.3.1 Data Acquisition and Temporal Edge Construction

For a market m, each trade is represented as $\tau = (w, t, p, s, o)$, where w is the wallet, t the timestamp, p the price, s the side (buy or sell) and o the outcome token. Trades are fetched with retrying requests, normalised and archived together with a manifest so that the same slice can be reproduced. Two trades by different wallets form a temporal co-trading edge when they occur within a window Δ:

$$e_{ij} = (w_i, w_j, t_i) \quad \text{if} \quad w_i \neq w_j \ \text{ and } \ |t_i - t_j| \leq \Delta \tag{1}$$

with $\Delta = 300$ seconds. To keep the graph tractable in busy periods, each trade is linked to at most 50 partners (fan-out limit). Edges include co-trades on opposite sides and opposite outcomes. For every unordered wallet pair (u, v) the edges are aggregated into the number of interaction events n_{uv}, the median absolute time difference, the median absolute price difference, the activity span, the number of distinct timestamps, and the proportions of same-side, same-outcome and same-behaviour events. In parallel, non-outcome wallet features describing activity, price, size and buy/sell behaviour are computed for use as node features.

3.3.2 Model-Free Lockstep Candidate Screen

A wallet pair is considered *tight* when it satisfies all of the following conditions:

$$n_{uv} \geq 10, \quad \text{med}|\Delta t| \leq 5 \text{ s}, \quad \text{med}|\Delta p| \leq 0.01 \tag{2}$$

Tight pairs are divided into two categories (Table 4). A pair is `persistent_tight` when its activity spans more than seven days and includes at least three distinct timestamps; otherwise it is `one_off_tight`. The label "one-off" does not mean a single trade; it means the tight behaviour does not persist over the required period. Same-side and same-outcome proportions are recorded but are not used as selection thresholds in the saved runs. The candidate sets therefore establish tightness in time and price, not necessarily same-side trading. No learned score is used at any point in the screen (invariant R1), which avoids the circularity of using a model to choose the pairs that are later used to validate it.

Table 4. Candidate selection rules applied by the lockstep screen

Rule	Condition	Purpose
Minimum events	$n_{uv} \geq 10$ co-trading events	Exclude incidental pairs
Time tightness	Median $\|\Delta t\| \leq 5$ seconds	Capture near-simultaneous trading
Price tightness	Median $\|\Delta p\| \leq 0.01$	Capture near-identical prices
persistent_tight	Span > 7 days and ≥ 3 distinct timestamps	Repeated behaviour over time
one_off_tight	Tight but not persistent	Concentrated tight behaviour
Recorded only	Same-side, same-outcome, same-behaviour shares	Descriptive; not thresholds

Co-timing permutation checks

To test whether tight pairs exceed what a busy market produces by chance, trades were shuffled within one-hour time bins (optionally also within side × outcome strata), the screen was recomputed, and the observed number of tight pairs was compared with $R = 20$ shuffles per mode.

A stricter *directional* definition additionally requires at least 80% of a pair's events to be on the same side and outcome. The empirical p-value uses the add-one estimator [25]:

$$p = (1 + \#\{\text{null} \geq \text{observed}\}) / (1 + R) \quad (3)$$

With 20 shuffles the smallest attainable value is $1/21 \approx 0.0476$.

3.3.3 On-Chain Funding Collection and Matched-Control Design

Transfer collection and truncation

Polygon native and ERC-20 transfers were collected for candidate wallets only. Some very active wallets reach the collector's row limit, which truncates their observed history. A wallet is flagged as truncated if it appears on the collector's cap list, or if it has at least 9,000 ERC-20 rows and its last observed ERC-20 timestamp is more than seven days before the end of the data. Truncated wallets are excluded in the sensitivity analysis. An unflagged wallet is "not identified as truncated", not proven to have a complete history.

Anchor-fixed matched controls

For each candidate pair (*A*, *B*), controls are formed by keeping one wallet fixed and replacing the other: (*A*, *C*) and (*B*, *D*) (Fig. 2). The partners *C* and *D* are chosen to have co-timing counts comparable to the candidate pair, are not themselves candidate pairs, and are not reused across controls. Fixing an anchor wallet holds that wallet's own funding propensity constant, so a very active wallet that transfers often contributes to both the candidate and the control arm. Candidate pairs without on-chain coverage, or touching an excluded wallet, are dropped and counted (invariant R3).

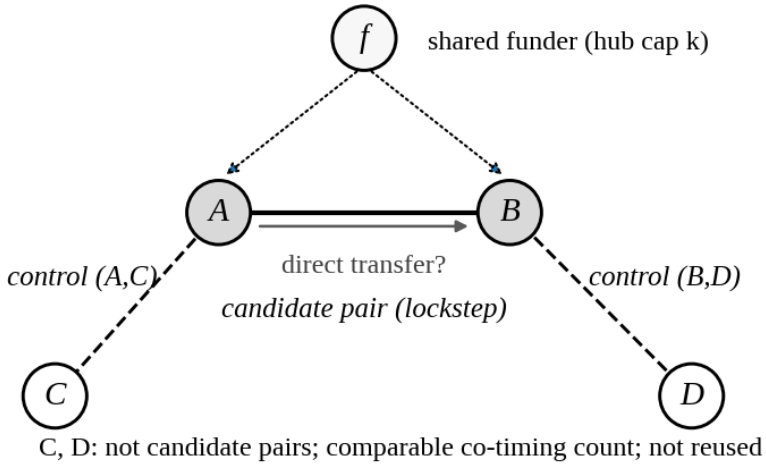


Figure 2. Anchor-fixed matched-control design and funding measures

Funding measures

The **primary measure** is whether either wallet transferred funds directly to the other at any time in the observed Polygon window. It does not depend on any hub threshold (invariant R4):

$$D(a, b) = 1[\ \exists \text{ transfer } a \to b \ \text{ or } \ b \to a\ ] \tag{4}$$

Secondary measures test whether the two wallets share a funder, while restricting funders by the number of fetched wallets they reach, $k \in \{5, 10, 20, 50, 100\}$:

$$SF_k(a, b) = 1[\ \exists f \notin \{a, b\} : f \to a, f \to b, |R(f)| \le k\ ] \tag{5}$$

where $R(f)$ is the set of fetched wallets funded by f. Deposit-only variants restrict shared funders to selected stablecoin deposits received no later than the wallet's first trade in the market, excluding the zero address. A rarity-weighted shared-deposit score gives more weight to uncommon funders. Table 5 summarises the measures. For each binary measure the effect is the difference between candidate and control rates, in percentage points:

$$\Delta = 100 \times (\ \text{mean}_{\text{candidates}}\ D - \text{mean}_{\text{controls}}\ D\ ) \tag{6}$$

Table 5. Funding measures used in the validation analysis

Measure	Definition	Role
Direct transfer	Either wallet sent funds to the other at any time in the window	Primary
Any-transfer shared funder	A third address funded both wallets and reaches $\le k$ fetched wallets	Secondary (5 caps)
Deposit-only shared funder	As above, restricted to stablecoin deposits before first market trade	Secondary (5 caps)
Rarity-weighted score	Shared deposit funders weighted towards uncommon funders	Secondary (raw units)

Wallet-level bootstrap

Pairs that share a wallet are not independent, so resampling pairs would understate uncertainty. Instead, wallets are reweighted [23]. In each of 2,000 bootstrap replicates, wallet weights are drawn from a multinomial distribution and each pair receives the product of its two wallets' weights:

$$\omega \sim \text{Multinomial}(n_W;\ 1/n_W, \ldots, 1/n_W), \quad w_{ab} = \omega_a \times \omega_b \tag{7}$$

Weighted rates are computed for both arms, and the 2.5th and 97.5th percentiles of the replicate differences form the 95% interval. These intervals describe sampling uncertainty conditional on the observed data and the matching procedure; they do not account for missing transfers and do not establish causality.

3.3.4 Memory-Based Temporal Model Architecture

The link-ranking model, implemented as `PolymapTGN`, is a custom memory-based temporal model (Fig. 3). A linear node encoder maps fit-period wallet features to a hidden representation. Each wallet keeps a memory vector that is updated after chronological interactions using a GRU-style cell [20]:

$$\mathbf{s}_i \leftarrow \mathrm{GRU}(\, \mathbf{s}_i, \mathbf{m}_i \,) \tag{8}$$

where $\mathbf{m}_i$ is the interaction message. A candidate link between wallets u and v is scored by the cosine similarity of the L2-normalised memories divided by a temperature $T = 0.1$, plus an optional pair-history term:

$$r(u, v) = \cos(\, \mathbf{s}_u, \mathbf{s}_v \,) / T \; + \; g_\psi(\, \boldsymbol{\varphi}_{uv} \,) \tag{9}$$

Here g_ψ is a small $6 \rightarrow 16 \rightarrow 1$ ReLU multilayer perceptron applied to six causal pair features:

$$\boldsymbol{\varphi}_{uv} = [\, \log(1+c),\, d_{5\mathrm{m}},\, d_{1\mathrm{h}},\, d_{24\mathrm{h}},\, 1[c > 0],\, \log(1+a) \,] \tag{10}$$

where c is the number of prior undirected co-trades of the pair, d_h is a count of prior co-trades exponentially decayed with half-life h (5 minutes, 1 hour and 24 hours), $1[c > 0]$ is a previously-seen flag, and a is the pair's age in hours. Pair state is undirected and is read *before* the current batch is absorbed, so no feature uses the event being predicted. The plain variant has no explicit pair history; the extension adds an explicit time-dependent pair signal.

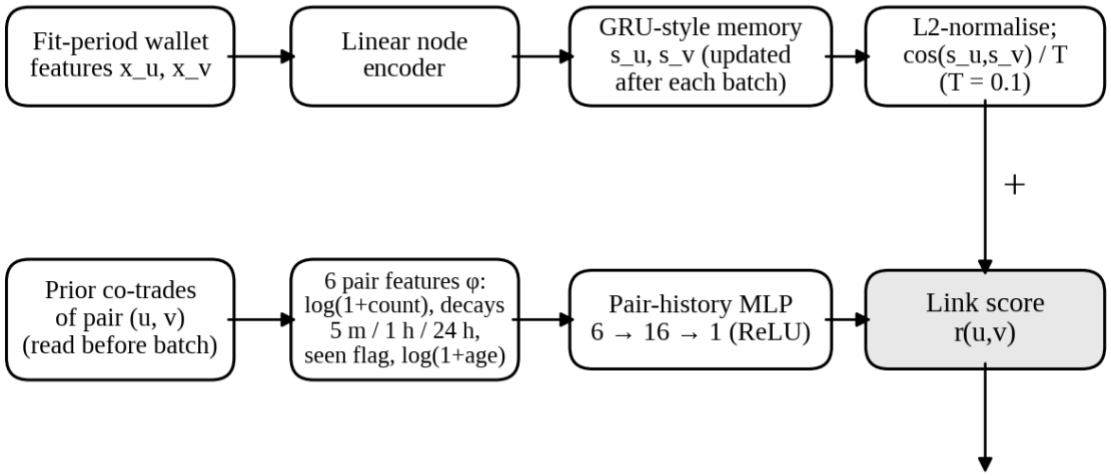


Figure 3. Architecture of the memory-based temporal link-ranking model

The model is inspired by TGN [10] but is not an implementation of TGN, TGAT [11] or JODIE [19]: it has no neighbour attention, no per-node time encoding, no last-update-time mechanism and no message store. It is therefore described as a memory-based temporal model. Three fixed variants were evaluated (Table 6).

Table 6. Fixed model variants evaluated in the final comparison

Variant	Pair history	Node memory	Learning rate	Epochs
Plain	Off	On	0.003	10
Tuned	On	On	0.003	20
Memory-off	On	Off	0.001	10

3.3.5 Dataset Description

Experiments were conducted on two geopolitical Polymarket markets that resolve on events "before 2027". Each run uses a 63-day trade slice, a 300-second edge window and a fan-out limit of 50 partners per trade. The observed trade windows (UTC) were:

- **Xi** (`xi_2027_63d_p50`): 2026-08-18 12:57:28 to 2026-10-20 12:33:41. Condition IDs are stored in `configs/markets/xi_2027_63d.yaml`.
- **Iran** (`rep_iran`): 2026-08-24 05:05:18 to 2026-10-26 04:34:14. Condition IDs are stored in `configs/markets/rep_iran.yaml`.

Table 7 summarises the trade graphs, candidate sets and on-chain data. Table 8 describes the candidate-wallet graphs and chronological splits used by the link-ranking model.

Table 7. Trade graphs, candidate sets and on-chain coverage

Item	Xi	Iran
Cached trades	23,839	35,135
Wallets in trade graph	2,461	3,761
Temporal edges	340,752	460,091
Candidate pairs (min-events = 10)	397	1,512
Persistent / one-off tight pairs	299 / 98	998 / 514
Candidate wallets	163	420
On-chain transfer rows	463,739	1,293,799
Candidate wallets with on-chain data	160 of 163	420 of 420
Wallets with truncated history	21 (comparable-rule list)	54
Candidate pairs involving truncated wallets	152 of 397	632 of 1,512

Table 8. Candidate-wallet model graphs and chronological splits

Item	Xi	Iran
Nodes (candidate wallets)	163	420
Edges in induced graph	78,306	280,656
Training events	54,813	196,076
Test events	23,493	84,580
Last training timestamp (UTC)	2026-09-29 19:25:09	2026-09-26 00:33:02
Test events involving a train-unseen wallet	3,205	16,205

The induced graph contains all temporal edges between candidate wallets, not only the edges of candidate pairs. The positive target in link ranking is a later observed co-trading edge, not a labelled financial relationship. Each market is trained separately; Iran reuses the settings, not weights learned on Xi.

3.3.6 Implementation Details

The pipeline is implemented as a Python package with layered configuration and disk-based stages invoked from a command-line interface. Table 9 lists the main components and Table 10 lists the command-line stages.

Table 9. Main software components

Component	Location	Use
Configuration	`configs/`, `config.py`	Market IDs, windows, thresholds, model settings, fingerprints
Trade client and cache	`data/client.py`, `data/trades.py`	Fetch normalised trades with retries; archive raw slices
Temporal edges	`features/edges.py`	Link wallets trading within 300 s
Wallet features	`features/wallets.py`	Activity, price, size and buy/sell node features
Graph bundle	`graph/build.py`	Standardised graph for downstream stages
Candidate screen	`analysis/relationships.py`	Model-free persistent/one-off tight pairs
Memory model	`models/temporal.py`	Node encoder, GRU-style memory, cosine score
Pair history	`models/pair_history.py`	Causal counts, decays, seen flag, pair age
Training and scoring	`training/trainer.py`	Chronological fit; scoring against sampled negatives
Negative samplers	`training/negatives.py`	Random, hard and popularity protocols
Baselines	`models/baselines.py`	Pair count, recency, popularity, common neighbours
Evaluation	`evaluation/metrics.py`	AUC, AP, MRR, Hits@k, tie-aware ranks
Null models	`analysis/null_model.py`	Shuffled-trade reference distributions

Component	Location	Use
On-chain collection	`collect_polygon_transfers.py`	Polygon transfers for candidate wallets
Truncation check	`find_truncated_wallets.py`	Manifest flags and row/time heuristic
Funding robustness	`analysis/funding_robustness.py`	Matched controls, cap sweeps, bootstrap
Experiment runners	`tune_tgnn.py`, `run_final_test.py`	Validation sweeps; fixed five-seed tests
Backend API and frontend	`api/`, `frontend/src/`	Serve artifacts; React/Vite wallet graph

Table 10. Command-line stage map

Stage	Input → output	Role in this study
`fetch`	Market config → cached trades, manifest	Archive the input sample
`build`	Trades → edges, wallet features, manifest	Construct the market graph
`candidates`	Edges → candidate pairs and wallet list	Select the model-free funding sample
`train`	Graph → metrics, event scores	Separate predictive experiment
`null`	Trades → shuffled-screen distributions	Compare with chance references
`funding-robustness`	Candidates, trades, edges, transfers → tables	Current funding analysis
`screen`, `validate-funding`	Legacy model-ranked view; legacy validator	Historical comparators only

Model training used the Adam optimiser with weight decay 0.00001, hidden dimension 32, training batch size 128 with one batch per optimiser step, and evaluation batch size 4. Node features were rebuilt from fit-period trades only; wallets unseen during training receive zero raw features before standardisation. Final runs used hard negatives and seeds 42, 43, 44, 45 and 46 for every variant in both markets, with no best-seed selection. A separate validation-only sweep on Xi held out the last 15% of the pre-test period.

3.3.7 Evaluation Metrics

The funding analysis reports candidate and control rates, their difference in percentage points, and 95% wallet-bootstrap intervals. The pre-agreed primary success criterion is that, in each market, the direct-transfer gap is positive **and** its 95% interval excludes zero. This criterion was recorded in the project notes before the Iran replication; Xi's primary measure was chosen after its initial analysis, so Iran is the planned replication test.

For link ranking, the chronological split targets 70% training and 30% test events, cutting between distinct timestamps. Scoring is online: each test batch is scored before its events update memory and pair history. For every positive event, 50 hard negatives are drawn from the source wallet's historical interaction pool. The main metric is mean reciprocal rank, computed with tie-aware ranks over the set of queries Q:

$$\mathrm{MRR} = (1 / |Q|) \, \Sigma_{q \in Q} \, 1 / \mathrm{rank}_q \tag{11}$$

Hits@10 is the fraction of queries for which the true partner is ranked in the top ten (ties handled in expectation). AUC measures how well scores separate positives from sampled negatives:

$$\mathrm{AUC} = \mathrm{P}(\, r(\mathrm{positive}) > r(\mathrm{negative}) \,) \tag{12}$$

MRR is a ranking metric, not a classification accuracy and not a probability of ownership. The metrics are conditional on the negative sampler. The **recency** baseline is a 24-hour exponentially decayed co-trade count updated online, analogous to time-windowed EdgeBank [12]. An earlier Xi benchmark also evaluated pair count, popularity and common-neighbour scorers [21] under hard and random negatives, with paired contiguous-block bootstrap intervals (1,000 draws, up to 50 blocks) [24]. Because evaluation negatives depend on the seed, the reported five-seed standard deviation reflects both training and negative-sampling variation and is not a confidence interval.

3.3.8 Flowchart / Algorithm

The complete experimental sequence is summarised in Fig. 4. The same steps are listed below; detailed pseudo-code for the screen, the matched-control bootstrap and the online evaluation is given in Appendix I.

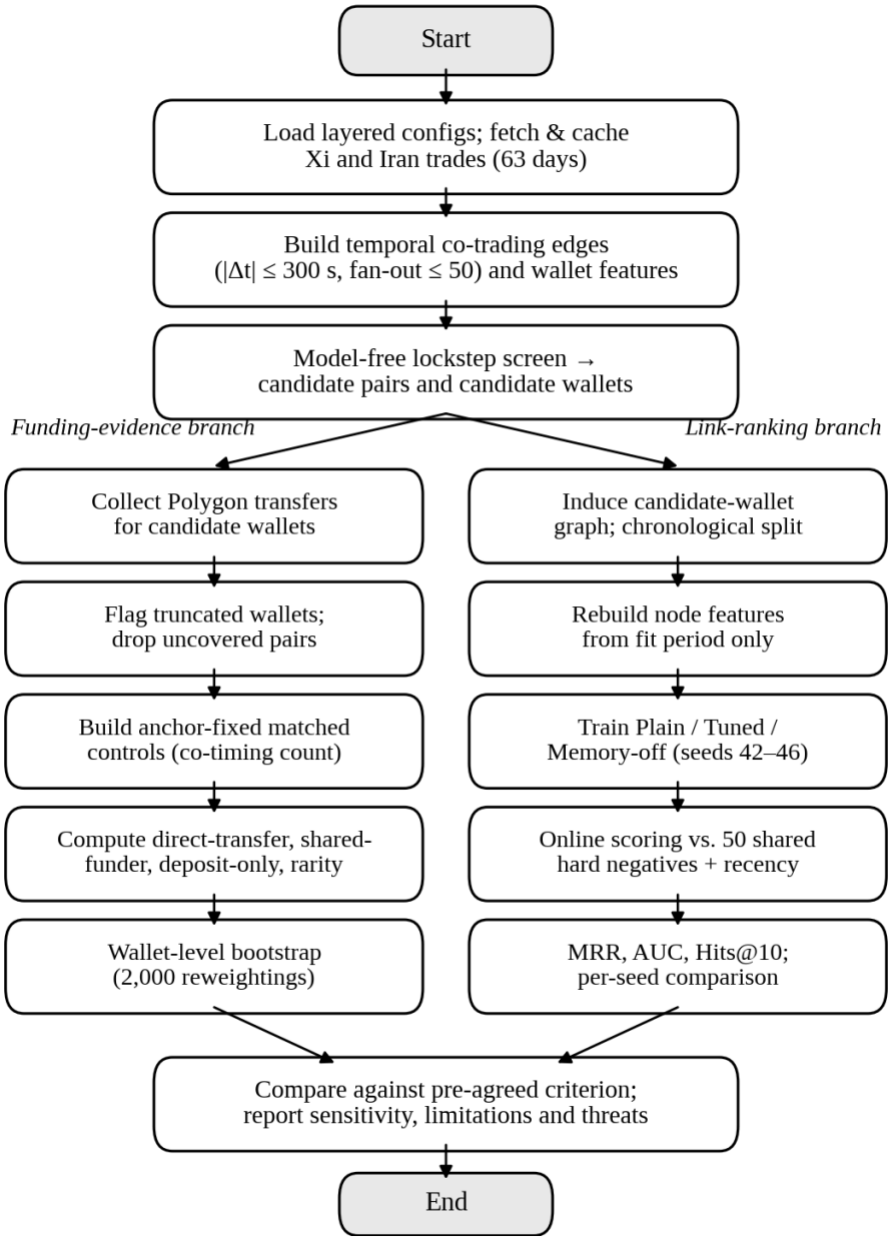


Figure 4. Overall experimental flowchart

1. Load the layered configuration and fetch and cache the Xi and Iran trade slices.
2. Build temporal co-trading edges (Δ = 300 s, fan-out $\leq$ 50) and fit-independent wallet features.
3. Apply the model-free lockstep screen to obtain candidate pairs and the candidate wallet list; run the permutation null on Xi.
4. Collect Polygon transfers for candidate wallets and flag truncated histories.

5. Build anchor-fixed matched controls, compute all funding measures and estimate wallet-bootstrap intervals for the primary (all-wallet) and sensitivity (truncation-excluded) analyses.
6. Induce the candidate-wallet graph, split it chronologically and rebuild node features from the fit period.
7. Train the plain, tuned and memory-off variants with five seeds, score online against shared hard negatives, and compare with the recency baseline.
8. Compare the results with the pre-agreed criterion and document sensitivity results, error patterns and limitations.

3.4 Summary

This chapter described the system architecture, computing environment, research invariants, edge construction, model-free lockstep screen, funding-validation design, memory-based temporal model, datasets, implementation details, evaluation metrics and the end-to-end algorithm. The design keeps candidate selection independent of the learned model and evaluates funding evidence against matched controls rather than against chance alone.

Chapter 4
Results and Analysis

This chapter presents the results of the co-timing permutation checks, the funding validation in both markets, and the separate link-ranking experiment. All numerical values are taken from the saved CSV and JSON artifacts, except the Xi primary funding table, whose original CSV was overwritten during later runs; its rounded values are taken from the preserved project write-up. All rate differences and intervals are in percentage points (pp); rarity-score rows are in raw score units. Rounded rates may not subtract exactly to the displayed gap.

4.1 Co-Timing Permutation Checks

The permutation checks were run on Xi only, using a separate edge-only screen with 12,172 eligible pairs and one-hour bins (Table 11). This screen differs from the final candidate manifest, which has 3,046 eligible pairs at min-events = 10, so the results are not a null test of the exact 397-pair funding sample. No corresponding Iran null model was run.

Table 11. Xi co-timing permutation checks (20 shuffles per mode)

Null stratification	Definition	Observed tight pairs	Null mean ± SD	Null range	Empirical p
Time bin	Directional	2,040	977.50 ± 62.99	831–1,074	0.0476
Time bin, side and outcome	Directional	2,040	1,958.05 ± 56.17	1,865–2,084	0.0952
Time bin	Timing/price only	3,343	3,521.00 ± 56.25	3,405–3,631	1.0000
Time bin, side and outcome	Timing/price only	3,343	3,285.70 ± 68.46	3,198–3,437	0.2381

Only the easiest null (time bins alone with the directional definition) reaches the minimum attainable p-value of 0.0476. The stricter directional null, which preserves each wallet's side and outcome preferences, gives p = 0.0952 and does not clear 0.05, and the timing/price-only definition is not distinguishable from chance. The co-timing checks therefore do not, on their own, establish pair-specific behaviour. This is the main reason why the funding analysis compares candidates with matched controls rather than with chance references.

4.2 Funding Validation: Xi Market

4.2.1 Primary analysis (all fetched wallets)

Of the 397 candidate pairs, 391 had on-chain coverage and were compared with 782 matched controls; six pairs were dropped because of missing coverage. Table 12 reports all measures.

Table 12. Xi primary funding results (all fetched wallets; 391 candidates, 782 controls)

Measure	Candidates	Controls	Gap (pp)	95% interval
Direct transfer, any time (primary)	15.3%	8.6%	+6.8	−0.3 to +16.7
Shared funder, any transfer, cap 5	24.8%	20.7%	+4.1	−7.3 to +15.7
Shared funder, any transfer, cap 10	38.1%	33.6%	+4.5	−8.9 to +16.6
Shared funder, any transfer, cap 20	50.6%	49.5%	+1.1	−12.6 to +12.3
Shared funder, any transfer, cap 50	58.8%	57.5%	+1.3	−12.5 to +12.7
Shared funder, any transfer, cap 100	85.7%	83.1%	+2.6	−8.3 to +11.2
Shared funder, deposit-only, cap 5	4.6%	4.3%	+0.3	−3.7 to +4.6
Shared funder, deposit-only, cap 10	8.7%	7.2%	+1.5	−4.8 to +8.0
Shared funder, deposit-only, cap 20	14.8%	12.3%	+2.6	−5.9 to +10.3
Shared funder, deposit-only, cap 50	19.2%	17.4%	+1.8	−6.9 to +10.2
Shared funder, deposit-only, cap 100	52.2%	45.9%	+6.3	−4.3 to +16.6
Rarity-weighted shared deposit funders	0.767	0.823	−0.056	−0.422 to +0.299

Source: section 6 of docs/funding_validation_findings.md (original primary run, one-decimal precision).

Candidates had a higher direct-transfer rate than controls (15.3% versus 8.6%), a gap of +6.8 points. However, the 95% interval runs from −0.3 to +16.7 and includes zero, so the primary criterion is **not met** for Xi. All shared-funder gaps are positive but small relative to their intervals, and the rarity score is slightly lower for candidates.

4.2.2 Sensitivity analysis (truncated wallets excluded)

The comparable sensitivity run excluded the 21 wallets flagged as truncated and used 242 candidate pairs and 484 controls (Table 13). In total 155 pairs were removed by the combined coverage and exclusion filter; 152 input pairs touch flagged wallets, and these overlap with the original six no-coverage pairs, so the counts cannot be added directly.

Table 13. Xi sensitivity funding results (21 truncated wallets excluded; 242 candidates, 484 controls)

Measure	Candidates	Controls	Gap (pp)	95% interval
Direct transfer, any time (primary)	14.05%	4.13%	**+9.92**	**+0.27 to +24.02**
Shared funder, any transfer, cap 5	21.90%	14.05%	+7.85	−7.13 to +22.33

Measure	Candidates	Controls	Gap (pp)	95% interval
Shared funder, any transfer, cap 10	34.71%	27.27%	+7.44	−9.64 to +22.76
Shared funder, any transfer, cap 20	49.17%	43.18%	+5.99	−12.14 to +21.33
Shared funder, any transfer, cap 50	54.13%	50.00%	+4.13	−15.50 to +20.64
Shared funder, any transfer, cap 100	79.75%	76.03%	+3.72	−12.22 to +17.05
Shared funder, deposit-only, cap 5	1.65%	0.41%	+1.24	−1.10 to +4.42
Shared funder, deposit-only, cap 10	4.96%	0.62%	+4.34	−0.42 to +11.83
Shared funder, deposit-only, cap 20	9.50%	3.93%	+5.58	−1.80 to +13.55
Shared funder, deposit-only, cap 50	10.33%	5.58%	+4.75	−3.92 to +13.46
Shared funder, deposit-only, cap 100	39.26%	29.34%	+9.92	−6.11 to +23.91
Rarity-weighted shared deposit funders	0.416	0.371	+0.046	−0.087 to +0.163

Source: data/artifacts/xi_2027_63d_p50/funding_robustness_table.csv. Bold: interval excludes zero.

After excluding truncated wallets the direct-transfer gap widens to +9.92 points, and its interval (+0.27 to +24.02) excludes zero, but only narrowly. The shared-funder rows are threshold checks on the same sample and should not be counted as independent discoveries; none of their intervals excludes zero.

4.3 Funding Validation: Iran Market

4.3.1 Primary analysis (all fetched wallets)

All 1,512 Iran candidate pairs had coverage and were compared with 3,023 controls (Table 14). Iran is the planned replication test for the pre-agreed criterion.

Table 14. Iran primary funding results (all fetched wallets; 1,512 candidates, 3,023 controls)

Measure	Candidates	Controls	Gap (pp)	95% interval
Direct transfer, any time (primary)	21.89%	17.30%	+4.59	−0.83 to +10.46
Shared funder, any transfer, cap 5	13.43%	15.22%	−1.79	−6.17 to +3.21
Shared funder, any transfer, cap 10	18.58%	22.00%	−3.41	−8.71 to +2.52
Shared funder, any transfer, cap 20	26.26%	28.35%	−2.09	−8.16 to +4.22
Shared funder, any transfer, cap 50	40.54%	43.33%	−2.79	−10.05 to +5.16
Shared funder, any transfer, cap 100	46.23%	51.01%	−4.78	−11.84 to +2.70
Shared funder, deposit-only, cap 5	1.26%	1.49%	−0.23	−1.32 to +1.04
Shared funder, deposit-only, cap 10	2.05%	2.45%	−0.40	−1.95 to +1.24
Shared funder, deposit-only, cap 20	5.16%	3.80%	+1.35	−1.28 to +4.81
Shared funder, deposit-only, cap 50	6.42%	5.43%	+0.99	−2.11 to +4.50
Shared funder, deposit-only, cap 100	8.00%	6.62%	+1.39	−2.08 to +5.15
Rarity-weighted shared deposit funders	0.435	0.474	−0.038	−0.177 to +0.062

Source: data/artifacts/rep_iran/funding_robustness_table_all_wallets.csv.

The direct-transfer gap is again positive (21.89% versus 17.30%, +4.59 points) but its interval (−0.83 to +10.46) includes zero. The pre-agreed criterion is therefore **not met in the replication market either**. Unlike Xi, the any-transfer shared-funder gaps are negative at every cap, so the broad shared-funder evidence does not replicate.

4.3.2 Sensitivity analysis (truncated wallets excluded)

The sensitivity run excluded the 54 flagged wallets and the 632 candidate pairs involving them, leaving 880 candidate pairs and 1,758 controls (Table 15).

Table 15. Iran sensitivity funding results (54 truncated wallets excluded; 880 candidates, 1,758 controls)

Measure	Candidates	Controls	Gap (pp)	95% interval
Direct transfer, any time (primary)	13.30%	7.74%	**+5.56**	**+0.16 to +12.26**
Shared funder, any transfer, cap 5	7.27%	6.77%	+0.50	−3.27 to +5.58
Shared funder, any transfer, cap 10	10.80%	11.72%	−0.92	−6.14 to +5.23
Shared funder, any transfer, cap 20	18.52%	16.72%	+1.80	−5.86 to +10.26
Shared funder, any transfer, cap 50	33.30%	31.74%	+1.55	−7.57 to +11.54
Shared funder, any transfer, cap 100	39.89%	40.96%	−1.07	−9.93 to +8.36
Shared funder, deposit-only, cap 5	0.34%	0.06%	+0.28	−0.21 to +1.36
Shared funder, deposit-only, cap 10	0.68%	0.23%	+0.45	−0.38 to +1.80
Shared funder, deposit-only, cap 20	3.98%	0.68%	**+3.29**	**+0.33 to +7.58**
Shared funder, deposit-only, cap 50	4.77%	1.25%	**+3.52**	**+0.51 to +7.80**
Shared funder, deposit-only, cap 100	5.91%	1.99%	**+3.92**	**+0.63 to +8.32**
Rarity-weighted shared deposit funders	0.291	0.273	+0.018	−0.041 to +0.081

Source: data/artifacts/rep_iran/funding_robustness_table.csv. Bold: interval excludes zero.

The direct-transfer gap increases to +5.56 points with an interval of +0.16 to +12.26, again excluding zero only narrowly. Iran's deposit-only rows at caps 20, 50 and 100 also exclude zero. These are correlated threshold variants of a single measure rather than three independent successes, and their Xi counterparts include zero. Unflagged wallets can still have unobserved transfers, so exclusion does not certify completeness.

4.4 Combined Funding Analysis

Table 16 and Fig. 5 bring together the primary measure across all four analyses.

Table 16. Direct-transfer gap across markets and analyses

Analysis	Xi	Iran
Primary (all wallets)	+6.8 pp, −0.3 to +16.7	+4.6 pp, −0.8 to +10.5
Truncation sensitivity	+9.9 pp, +0.3 to +24.0	+5.6 pp, +0.2 to +12.3
Pre-agreed criterion met?	No	No (planned replication)

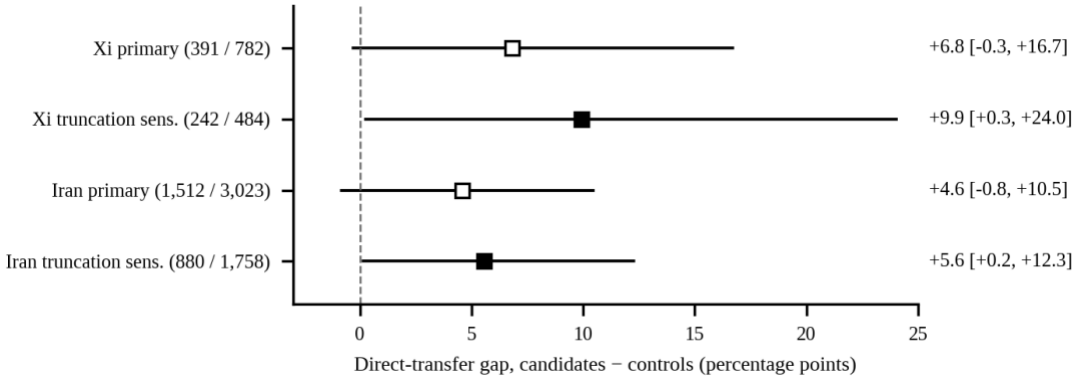


Figure 5. Direct-transfer gap with 95% wallet-bootstrap intervals (filled markers exclude zero)

The direction replicates across both markets and both analyses, but the primary criterion (same direction with an interval excluding zero) is not met in either market. The sensitivity intervals exclude zero only by a fraction of a percentage point, and they describe reduced samples from which 39% of Xi and 42% of Iran candidate pairs, disproportionately the heavier wallets, were removed. Fig. 6 shows that the shared-funder measures behave differently: the any-transfer gaps are imprecise on Xi and mostly negative on Iran, while the deposit-only gaps are small and only the Iran sensitivity rows at the larger caps exclude zero. The rarity score shows no consistent signal. The overall funding result is therefore **suggestive, not confirmed**.

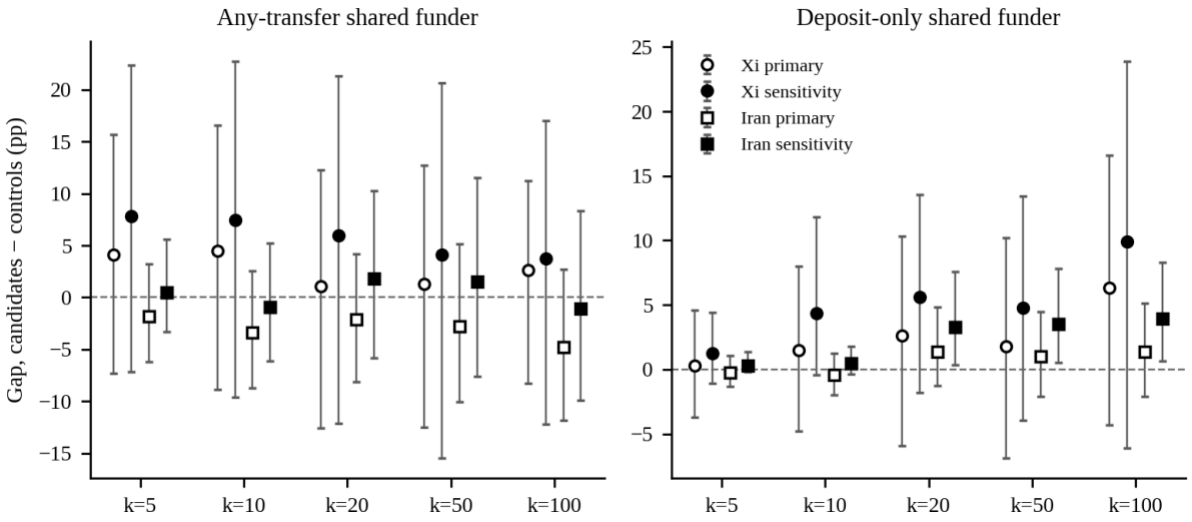


Figure 6. Shared-funder gaps across hub caps for both markets and analyses

4.4.1 Historical funding results retained for context

Several earlier analyses were superseded during the project. They are retained in Table 17 because they explain how the final design evolved, but they are not additional replications.

Table 17. Superseded and historical funding analyses

Analysis	Result	Status
Xi 16-wallet exclusion	253 candidates, 506 controls; direct transfer 13.4% vs 4.2%, gap +9.3 [+0.6, +23.1]	Superseded by 21-wallet list
Legacy Xi validator	397 candidates, 1,090 controls; direct 15.11% vs 7.16% (+7.96); any-link 98.99% vs 98.81%, permutation p = 0.800	Hub saturation; weaker matching
Discarded xi_2027 comparison	2,100-trade slice paired with 63-day transfers; 613 candidates, shared funder 41.6% vs 100%	Invalid: unequal coverage
Post-hoc pooling	Primary gaps about +5.3 [+0.6, +10.0]; sensitivity gaps about +6.5 [+1.1, +11.9]	Not prespecified; does not rescue criterion

The legacy validator is particularly instructive: almost every wallet shared some funder once hub addresses were admitted (98.99% versus 98.81%), so unrestricted shared-funder measures have essentially no discriminating power. This motivated the hub-cap sweep, the deposit-only restriction and the anchor-fixed matching used in the final design. The post-hoc pooled estimates were rough inverse-variance combinations inferred from rounded, asymmetric intervals under an independence assumption; they are not a raw-data meta-analysis.

4.5 Temporal Link Ranking

4.5.1 Validation and test-use history

A local validation-only sweep on Xi held out 15% of the pre-test period (46,593 fit events and 8,220 validation events) using train-period features, hard negatives, hidden dimension 32, learning rate 0.003 and seeds 42–44 (Table 18).

Table 18. Local Xi validation sweep (hard negatives)

Variant	Epochs	MRR seed 42	MRR seed 43	MRR seed 44	Mean MRR ± SD	Mean AUC	Mean Hits@10
Pair history on	10	0.467	0.284	0.279	0.343 ± 0.107	0.567	0.462
Pair history on	20	0.305	0.276	0.275	0.285 ± 0.017	0.469	0.381
Pair history off	10	0.423	0.436	0.436	**0.432 ± 0.008**	**0.731**	**0.555**

The 20-epoch pair-history configuration that became the "tuned" variant **was not** the best local validation result. It was carried forward from an earlier sandbox experiment whose code and raw files were not available locally (Table 19). The discrepancy between the sandbox and local sweeps remains unresolved, and the reconstructed implementation cannot be claimed to reproduce the sandbox study.

Table 19. Earlier sandbox validation results (reported, not locally verified)

Sandbox variant	Mean validation MRR
Defaults, original full-period features	0.177
Defaults, train-period features	0.173
Train-period features, 10 epochs, no pair history	0.334
Pair history, 10 epochs	0.589
Pair history, 20 epochs, learning rate 0.003	0.616
Memory off, pair history, 10 epochs	0.583
Recency reference	0.469

The Xi test period was also not used only once. Earlier baseline runs had already evaluated on it, and after the local sweep a one-seed tuned evaluation reported MRR 0.577 and AUC 0.862, followed by the saved five-seed run (seed 42 MRR 0.600). No saved artifact explains the same-seed variation. The Xi result should therefore be called exploratory rather than a single-use holdout [28]. Iran used the fixed runner without any local tuning.

4.5.2 Five-seed results: Xi

Table 20. Xi five-seed test results (23,493 test events; recency MRR 0.433)

Variant	MRR mean	MRR SD	AUC mean	Hits@10 mean	MRR gap vs recency
Plain	0.341	0.019	0.689	0.481	−0.093
Tuned	**0.586**	0.015	**0.865**	**0.804**	**+0.153**
Memory-off	0.182	0.033	0.563	0.330	−0.252

Table 21. Xi per-seed MRR and AUC

Variant	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
Plain MRR	0.335	0.321	0.343	0.371	0.334
Plain AUC	0.691	0.678	0.699	0.680	0.695
Tuned MRR	0.600	0.590	0.594	0.586	0.561
Tuned AUC	0.867	0.870	0.865	0.857	0.865
Memory-off MRR	0.182	0.149	0.184	0.234	0.159
Memory-off AUC	0.548	0.531	0.587	0.616	0.534

On Xi, the tuned configuration achieved a mean test MRR of 0.586 with a small seed standard deviation of 0.015, mean AUC 0.865 and Hits@10 0.804. It beat the recency baseline (0.433) in all five seeds, with a mean gap of +0.153. The plain configuration (0.341) and the memory-off configuration (0.182) were both below recency.

4.5.3 Five-seed results: Iran

Table 22. Iran five-seed test results (84,580 test events; recency MRR 0.330)

Variant	MRR mean	MRR SD	AUC mean	Hits@10 mean	MRR gap vs recency
Plain	0.156	0.012	0.587	0.308	−0.174
Tuned	**0.321**	0.056	**0.680**	**0.498**	−0.009
Memory-off	0.161	0.010	0.552	0.300	−0.170

Table 23. Iran per-seed MRR and AUC

Variant	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
Plain MRR	0.145	0.174	0.150	0.150	0.161
Plain AUC	0.580	0.610	0.592	0.573	0.578
Tuned MRR	0.251	0.366	0.319	0.387	0.285
Tuned AUC	0.615	0.707	0.687	0.730	0.663

Variant	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
Memory-off MRR	0.144	0.170	0.163	0.168	0.159
Memory-off AUC	0.526	0.570	0.564	0.552	0.548

On Iran, the tuned configuration again improved strongly over the plain variant (0.321 versus 0.156) but its mean MRR was slightly below recency (gap −0.009). It beat recency in only two of five seeds (seeds 43 and 45), and its seed-to-seed variation was much larger (MRR 0.251 to 0.387, SD 0.056). This supports describing the tuned model as "competitive with recency" on Iran; it is not a statistical equivalence or superiority result.

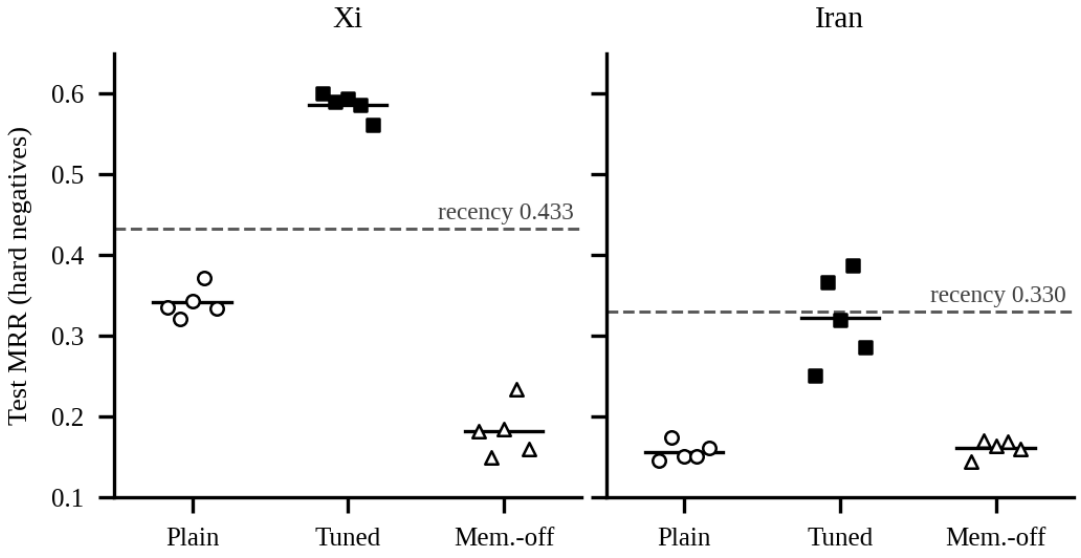


Figure 7. Per-seed test MRR and five-seed means for each variant compared with the recency baseline

4.5.4 Interpretation of model variants

The tuned model improves over the plain model by +0.246 MRR on Xi and +0.166 on Iran (Fig. 7). However, these comparisons are not clean ablations. Tuned versus plain changes both the pair features and the number of epochs. The memory-off variant additionally changes the learning rate and epochs and retains the static node-encoder cosine term, so it is not a pure pair-history-only model. In addition, with memory updates disabled the trainer still detaches memory after optimiser steps and does not refresh static embeddings within the epoch, which makes this branch unsuitable as a definitive memory ablation without further verification. The saved numbers are reported as run. Consequently, the results cannot isolate the benefit of pair history or prove that node memory is essential; earlier conversational claims that "memory still matters" were stronger than the

experiment supports. Differences in graph density, wallet universe and temporal patterns are plausible explanations for the gap between markets but were not tested.

4.5.5 Earlier Xi baseline benchmark

Before the frozen five-seed comparison, the original (plain) model and four non-learned scorers were evaluated on the same Xi test events under hard and random negatives (Table 24, Table 25 and Fig. 8). Pair count counts prior undirected co-trades; recency weights them by age; popularity ranks the target's prior event degree; common neighbours counts shared prior co-trading partners.

Table 24. Earlier Xi benchmark: MRR and AUC by negative sampler

Scorer	Hard MRR	Hard AUC	Random MRR	Random AUC
Original memory model	0.336	0.712	0.370	0.787
Pair count	0.223	0.543	0.431	0.855
Recency	**0.433**	**0.809**	**0.585**	**0.923**
Popularity	0.142	0.486	0.295	0.814
Common neighbours	0.112	0.456	0.273	0.675

Table 25. Earlier Xi benchmark: additional metrics and paired MRR gaps

Sampler	Scorer	AP	Hits@1	Hits@5	Hits@10	Model minus scorer MRR [95% CI]
Hard	Original model	0.140	0.252	0.390	0.507	—
Hard	Pair count	0.022	0.170	0.229	0.294	+0.113 [+0.061, +0.159]
Hard	Recency	0.109	0.344	0.503	0.643	−0.098 [−0.146, −0.053]
Hard	Popularity	0.020	0.088	0.142	0.208	+0.193 [+0.139, +0.250]
Hard	Common neighbours	0.018	0.043	0.136	0.215	+0.223 [+0.162, +0.295]
Random	Original model	0.199	0.263	0.457	0.595	—
Random	Pair count	0.141	0.266	0.643	0.833	−0.061 [−0.104, −0.022]
Random	Recency	0.364	0.428	0.811	0.923	−0.215 [−0.255, −0.176]
Random	Popularity	0.090	0.144	0.456	0.697	+0.074 [+0.013, +0.140]
Random	Common neighbours	0.054	0.124	0.448	0.603	+0.096 [+0.027, +0.171]

Intervals are paired contiguous-event-block bootstrap intervals for one trained model (1,000 draws, up to 50 blocks), not five-seed intervals.

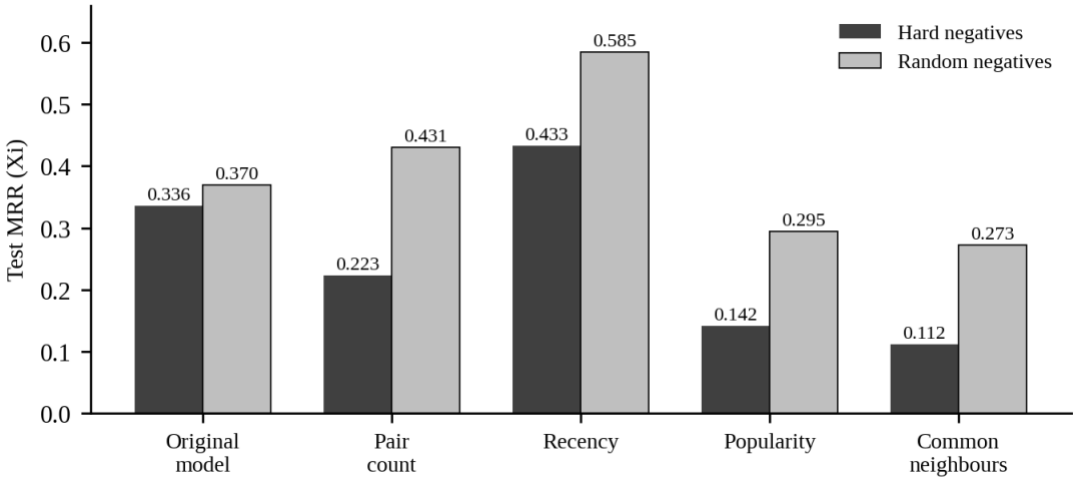


Figure 8. Earlier Xi benchmark: test MRR of the original model and non-learned scorers under hard and random negatives

Under hard negatives, recency beat the original model by 0.098 MRR, while the model beat pair count, popularity and common neighbours. Under random negatives even the pair-count rule beat the model. The choice of negative sampler therefore changes which rule is advantaged, consistent with the findings of Poursafaei *et al.* [12]. Because hard negatives are chosen from contact history, they favour scorers that use recency of contact. The final two-market runner evaluates only recency as the non-learned reference, so it does not establish the tuned model's superiority over all four baselines on Iran or under all samplers.

4.6 Error Patterns and Interpretation

Table 26 organises the main error sources and threats observed during the diagnostic runs into six categories.

Table 26. Error and threat taxonomy with mitigations

No.	Category	Trigger	Mitigation / status
1	Coverage truncation	Heavy wallets hit collector caps (152 Xi and 632 Iran pairs affected)	Sensitivity exclusion; unflagged ≠ complete
2	Hub funders	Exchanges, bridges and routers fund many wallets (legacy any-link 98.99% vs 98.81%)	Cap sweep, deposit-only restriction, rarity weighting
3	Retrospective wallet universe	Candidate wallets chosen from the full 63 days before the split	Fit-period features; disclosed as limitation
4	Test-set reuse	Xi test inspected by earlier and one-seed runs	Labelled exploratory; Iran run fixed
5	Confounded ablations	Variants change epochs and learning rate together with components	Reported as configurations, not components
6	Multiplicity	Eleven secondary rows per run with no correction	Treated as threshold checks [27]

Truncation. The sensitivity runs remove roughly two-fifths of the candidate pairs and disproportionately remove heavy wallets. The gap grows after exclusion in both markets. This is consistent with truncation hiding transfers in heavy wallets, but it is equally consistent with lighter wallets behaving differently; the design cannot distinguish these explanations.

Hub funders. The near-100% shared-funder rates in the legacy validator show that unrestricted common-funder heuristics have almost no discriminating power on Polygon. Even with caps, the any-transfer funder gaps did not replicate, while the hub-independent direct-transfer measure carried the only consistent signal.

Retrospective universe and test reuse. Selecting candidate wallets from the full 63-day window fixes the model's wallet universe before the train/test split; standardisation also uses this fixed universe, including zero-filled future-only wallets. Train-period feature rebuilding removes future-trade aggregation from node features but does not make the wallet universe prospective. Combined with the repeated inspection of the Xi test period, this limits the model results to exploratory link ranking.

Implementation details. Batch order is a chronological processing order rather than strictly earlier wall-clock time: equal-timestamp events can fall in different batches, so previously processed simultaneous events can affect later scores. Training does not reject source/true-target negative collisions; evaluation redraws collisions up to ten times but does not enforce zero remaining collisions; and hard-negative pools are built from the full fit period rather than updated before each training event. These details should be audited before any stronger prospective claim.

4.7 Summary

The permutation checks did not establish pair-specific co-timing on their own. Lockstep pairs showed higher direct-transfer rates than anchor-matched controls in both markets, but neither primary interval excluded zero, so the pre-agreed criterion was not met; the truncation-sensitivity intervals excluded zero only narrowly. Shared-funder evidence did not replicate. The tuned temporal model improved strongly over its plain variant in both markets, beat recency on Xi in all seeds, and was competitive with recency on Iran. The overall result is suggestive, not confirmed.

Chapter 5
Advantages, Limitations and Applications

5.1 Advantages

The proposed framework has the following advantages:

1. The candidate screen is model-free and uses explicit, interpretable thresholds, which avoids circularity between candidate selection and model validation.
2. Anchor-fixed matching on co-timing intensity provides a meaningful reference group, so funding links are judged relative to comparable wallet pairs rather than to chance alone.
3. The primary funding measure (direct transfer) does not depend on any hub threshold, and secondary measures expose how sensitive shared-funder evidence is to infrastructure addresses.
4. Wallet-level bootstrap intervals account for the dependence created when many pairs share the same wallet.
5. The link-ranking evaluation is online and uses exact shared hard negatives, so the learned model and the non-learned baselines are compared on identical candidates.
6. The pipeline is modular, configuration-driven and reproducible at the level of commands and inputs, with saved manifests and build fingerprints.
7. Results are reported against a pre-agreed criterion, including negative results, test-set reuse and confounded comparisons, which makes the evidence easy to audit.

5.2 Limitations

The present study has the following limitations:

1. The primary funding intervals include zero in both markets, so the pre-agreed criterion was not met; the sensitivity results rely on reduced, lighter-wallet samples.
2. Funding before the observed window is invisible, Polygon-only data omit other chains and exchange-internal transfers, and unflagged wallets may still have incomplete histories.
3. Exchanges, bridges, routers, custodians and protocol contracts can create funding links without common ownership; observed links are not proof of a relationship.

4. The secondary funding measures were not corrected for multiplicity, and only two markets, both geopolitical “before 2027” markets, were studied.
5. Candidate thresholds and the 300-second interaction window are research choices that change the sample; same-side trading is recorded but not required.
6. The model variant comparisons change several settings at once, the memory-off branch has an implementation caveat, the Xi test period was accessed more than once, the chosen configuration was not the best local validation result, and earlier sandbox results were not reproduced.
7. The candidate wallet universe was selected from the full period before the split, so the link-ranking evaluation is retrospective rather than prospective.
8. The final runner did not save model weights, full negative matrices or an experiment manifest, and Xi’s primary funding CSV was overwritten, so byte-for-byte reproduction has not been established.

5.3 Applications

After further validation, the framework may support the following applications:

1. Market-integrity monitoring for prediction-market operators, by flagging lockstep wallet pairs for human review alongside counterparty-based wash-trading detectors.
2. Research on coordinated behaviour in on-chain markets, using matched controls to separate pair-specific funding links from infrastructure effects.
3. Triage support for analysts, using the link-ranking model to prioritise which wallet pairs are likely to co-trade next.
4. A teaching and research baseline for honest evaluation of temporal graph models against memorisation heuristics under hard-negative protocols.
5. An evidence-oriented wallet-graph visualisation that displays lockstep edges, observed funding edges, shared funders and coverage status with explicit, non-accusatory labels.

5.4 Summary

The framework provides an auditable basis for studying lockstep wallets and their funding links, but its results must be interpreted within the limits of two markets, Polygon-only data, truncated

histories and exploratory model evaluation. The next stage is a prospective, preregistered test on new markets with complete transfer histories.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

This project developed Polymap, an exploratory pipeline for discovering lockstep wallet pairs on Polymarket and validating them with on-chain funding evidence, together with a separate memory-based temporal link-ranking experiment. On two 63-day geopolitical markets (Xi: 23,839 trades, 397 candidate pairs; Iran: 35,135 trades, 1,512 candidate pairs), lockstep pairs showed higher direct-transfer rates than anchor-matched controls in both markets, with gaps of +6.8 and +4.6 percentage points. Neither primary 95% interval excluded zero, so the pre-agreed success criterion was not met. Excluding wallets with truncated transfer histories produced gaps of +9.9 and +5.6 points whose intervals excluded zero only narrowly, and shared-funder evidence did not replicate across markets. The tuned memory-based model with causal pair history improved held-out co-trading-link ranking over its plain variant in both markets (MRR 0.586 versus 0.341 on Xi; 0.321 versus 0.156 on Iran), beat the recency baseline on Xi in every seed, and reached a competitive level on Iran. The appropriate conclusion is that tight lockstep pairs show a suggestive but unconfirmed excess of observed direct transfers relative to matched controls, and that a tuned memory-based model improves held-out co-trading-link ranking, strongly on Xi and to a competitive level on Iran. Neither result establishes common ownership, coordination or intent.

6.2 Future Scope

The future scope of this work includes the following directions:

1. Run a prospective, externally preregistered test on new markets with a single-use holdout period, so that both the funding criterion and the model comparison are evaluated without test-set reuse.
2. Recover complete transfer histories for heavy wallets, add other chains and exchange-aware funding data, and maintain a list of known infrastructure addresses to reduce hub confounding.
3. Add same-side and same-outcome selection thresholds and test sensitivity to the 300-second window and the candidate thresholds.

4. Perform clean single-factor ablations of pair history and node memory, fix the memory-off implementation caveat, and report event-level bootstrap intervals for ranking metrics.
5. Compare against published temporal graph models (TGN, TGAT) and EdgeBank using TGB-style evaluation, and save model checkpoints, negative matrices and experiment manifests.
6. Compare lockstep pairs with counterparty-based wash-trading clusters [4] to understand how the two behavioural signatures overlap.
7. Implement the proposed static research-data export and React visualisation with five layers: candidate wallet nodes, undirected lockstep edges, directed funding edges from actual transfer records, shared-funder nodes filtered by hub cap, and an evidence panel with rates, intervals, coverage and the exploratory label. Excluded or uncovered wallets must be shown as having incomplete coverage, not as having no funding link.

6.3 Summary

This chapter consolidated the conclusions from the funding validation and link-ranking experiments and defined a future research path focused on prospective validation, complete funding data, clean ablations, stronger baselines and an evidence-oriented visualisation.

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Appendix I

Software Flowchart and Algorithm

This appendix records the software sequence used in the experiments. The flowchart is provided in Chapter 3 as Figure 4; the three core procedures are reproduced below as pseudo-code.

Algorithm 1: Model-free lockstep screen

```
Require: trades T, window Δ = 300 s, fan-out F = 50, minimum events n0 = 10
Ensure : candidate pairs P with category labels
 1: E ← ∅
 2: for all trades τi in time order do
 3:     Ni ← up to F trades τj with wj ≠ wi and |ti − tj| ≤ Δ
 4:     E ← E ∪ {(wi, wj, ti, |Δt|, |Δp|) : τj ∈ Ni}
 5: end for
 6: for all unordered wallet pairs (u, v) in E do
 7:     compute n_uv, median|Δt|, median|Δp|, span, distinct timestamps
 8:     if n_uv ≥ n0 and median|Δt| ≤ 5 s and median|Δp| ≤ 0.01 then
 9:         if span > 7 days and distinct ≥ 3 then label persistent_tight
10:         else label one_off_tight
11:         P ← P ∪ {(u, v)}
12:     end if
13: end for
14: return P            ▷ no learned model score is used
```

Algorithm 2: Anchor-fixed matched controls and wallet-level bootstrap

```
Require: candidates P, co-timing counts n, transfers X, excluded wallets Z, B = 2000
Ensure : gap Δ and 95% interval for each funding measure
 1: C ← ∅;  U ← ∅                          ▷ U = control partners already used
 2: for all (A, B) ∈ P do
 3:     if A ∈ Z or B ∈ Z or pair lacks coverage then drop and count; continue
 4:     for anchor a ∈ {A, B} do
 5:         c ← partner with n_ac ≈ n_AB, (a, c) ∉ P, c ∉ U, covered
 6:         if c exists then C ← C ∪ {(a, c)};  U ← U ∪ {c}
 7:     end for
 8: end for
 9: compute direct transfer, SF_k, deposit-only and rarity for P and C
10: for b = 1 to B do
11:     draw wallet weights ω ~ Multinomial(nW; 1/nW, ..., 1/nW)
12:     pair weight w_ab ← ω_a × ω_b
13:     Δ_b ← weighted rate(P) − weighted rate(C)
14: end for
15: return Δ, [Q_0.025(Δ_b), Q_0.975(Δ_b)]
```

Algorithm 3: Online scoring with shared hard negatives

```
Require: test batches B1..BK (size 4), memories S, pair history H, scorers {model,
recency}
Ensure : MRR, Hits@10 and AUC for each scorer
 1: for k = 1 to K do
 2:     for all events (u, v, t) ∈ Bk do
 3:         N ← 50 hard negatives from u's historical pool (≤ 10 collision redraws)
 4:         score v and N with every scorer using the same N
 5:         record tie-aware rank of v for each scorer
 6:     end for
 7:     update memories S (GRU) and pair history H with events of Bk   ▷ after scoring
 8: end for
 9: return MRR, Hits@10, AUC
```

Appendix II

Dataset and Model Specifications

The following summary consolidates the datasets and model configuration described in the report.

- **Markets:** Xi (`xi_2027_63d_p50`) and Iran (`rep_iran`); 63-day trade slices; condition IDs in `configs/markets/xi_2027_63d.yaml` and `configs/markets/rep_iran.yaml`. The p50 file is an overlay, so Xi also requires the base 63-day market configuration.
- **Edge construction:** co-trading window $\Delta = 300$ s; maximum fan-out 50 partners per trade; opposite-side and opposite-outcome co-trades included.
- **Candidate screen:** min-events = 10; median $|\Delta t| \leq 5$ s; median $|\Delta p| \leq 0.01$; persistent if span > 7 days and ≥ 3 distinct timestamps.
- **Truncation rule:** collector cap list, or $\geq 9{,}000$ ERC-20 rows with last ERC-20 timestamp > 7 days before data end. Exclusion files: `truncated_xi_v2.csv` (21 wallets) and `truncated_rep_iran.csv` (54 wallets).
- **Funding analysis:** two controls per candidate where available; funder caps $k \in \{5, 10, 20, 50, 100\}$; 2,000 wallet-level bootstrap reweightings.
- **Model:** linear node encoder; GRU-style memory; cosine score with temperature 0.1; optional $6 \rightarrow 16 \rightarrow 1$ ReLU pair-history MLP; hidden dimension 32; Adam; weight decay 0.00001; training batch 128; evaluation batch 4; CPU.
- **Variants:** plain (no pair history, lr 0.003, 10 epochs); tuned (pair history, lr 0.003, 20 epochs); memory-off (pair history, no memory updates, lr 0.001, 10 epochs).
- **Evaluation:** 70/30 chronological split between distinct timestamps; 50 hard negatives; seeds 42–46; metrics MRR, AUC, Hits@10; recency baseline = 24-hour exponentially decayed co-trade count.
- **Fingerprints:** build `2412c5350ad0` (Xi) and `b00535329e18` (Iran); candidate manifests `f4aebcdf4606` (Xi) and `f9491d23ddda` (Iran). These are build fingerprints, not hashes of trained models.

Appendix III

Source Code and Sample Code

The principal reproducibility files are listed below; the full source is held in the project repository [29]. Commands are run from the `backend/` directory.

- `src/polymap/features/edges.py` – temporal co-trading edge construction.
- `src/polymap/analysis/relationships.py` – model-free lockstep candidate screen.
- `src/polymap/analysis/null_model.py` – co-timing permutation checks.
- `collect_polygon_transfers.py`, `find_truncated_wallets.py`, `recover_erc20.py` – on-chain collection, truncation checks and recovery.
- `src/polymap/analysis/funding_robustness.py` – matched controls, funder-cap sweeps and wallet bootstrap.
- `src/polymap/models/temporal.py`, `src/polymap/models/pair_history.py` – memory-based model and causal pair features.
- `src/polymap/training/trainer.py`, `negatives.py`, `splits.py` – training, negative sampling and chronological splitting.
- `src/polymap/models/baselines.py`, `src/polymap/evaluation/metrics.py` – non-learned baselines and metrics.
- `tune_tgnn.py`, `run_final_test.py` – validation sweeps and fixed five-seed test runner.

Build and select model-free candidates (Xi):

```
.venv/bin/polymap build -c configs/default.yaml \
  -c configs/markets/xi_2027_63d.yaml -c configs/markets/xi_2027_63d_p50.yaml \
  --run-name xi_2027_63d_p50
.venv/bin/polymap candidates -c configs/default.yaml \
  -c configs/markets/xi_2027_63d.yaml -c configs/markets/xi_2027_63d_p50.yaml \
  --run-name xi_2027_63d_p50 --min-events 10
```

Funding robustness, primary then sensitivity (Iran):

```
.venv/bin/polymap funding-robustness -c configs/default.yaml \
  -c configs/markets/rep_iran.yaml --run-name rep_iran \
  --funding-file data/raw/onchain_rep_iran/polygon_transfers.parquet
cp -n data/artifacts/rep_iran/funding_robustness_table.csv \
  data/artifacts/rep_iran/funding_robustness_table_all_wallets.csv
.venv/bin/polymap funding-robustness -c configs/default.yaml \
  -c configs/markets/rep_iran.yaml --run-name rep_iran \
  --funding-file data/raw/onchain_rep_iran/polygon_transfers.parquet \
```

```
  --exclude-wallets-file truncated_rep_iran.csv
```

Five-seed final model comparison:

```
.venv/bin/python run_final_test.py -c configs/default.yaml \
  -c configs/markets/xi_2027.yaml --run-name xi_2027_63d_p50
.venv/bin/python run_final_test.py -c configs/default.yaml \
  -c configs/markets/rep_iran.yaml --run-name rep_iran
```

The Xi model command used `xi_2027.yaml` but read the already-built p50 graph from the overridden run directory; it must not be mistaken for the graph-build recipe. Rerunning with default output paths replaces the saved final CSVs, so a separately named reproduction should use the `--output` argument.]

Appendix IV

Raw Experimental Observations

The following observations are reported in Chapter 4. They should be retained together with the original CSV/JSON artifacts when the final submission package is prepared.

- **Xi funding:** primary direct transfer 15.3% vs 8.6%, gap +6.8 [−0.3, +16.7]; sensitivity 14.05% vs 4.13%, gap +9.92 [+0.27, +24.02].
- **Iran funding:** primary direct transfer 21.89% vs 17.30%, gap +4.59 [−0.83, +10.46]; sensitivity 13.30% vs 7.74%, gap +5.56 [+0.16, +12.26].
- **Xi null:** directional time-bin p = 0.0476; directional side/outcome p = 0.0952; timing/price p = 1.0000 and 0.2381.
- **Xi model:** tuned MRR 0.586 ± 0.015 (AUC 0.865, Hits@10 0.804); recency 0.433; tuned beat recency in 5/5 seeds.
- **Iran model:** tuned MRR 0.321 ± 0.056 (AUC 0.680, Hits@10 0.498); recency 0.330; tuned beat recency in 2/5 seeds.
- **One-seed Xi session output:** tuned MRR 0.577, AUC 0.862 (not saved as an artifact).

Earlier engineering measurements (not re-benchmarked): training on 100,000 edges for one epoch changed from 94 seconds to about 4 seconds after batching; a 340,000-edge, three-epoch, three-sampler run took 76 seconds; candidate-subgraph training took about 30 seconds. The older full-run estimate of more than 20 minutes was extrapolated. Hardware, epoch counts and the pair-feature path differ from the final experiment, so these are not estimates for the 15-run final comparison.

Verification status: earlier session checks recorded 57 passing local tests, syntax-check success, and targeted import and logging lint success; the broad repository lint check still had unrelated findings. Both 15-row final-result CSVs were reconciled with their summaries, and 77 report rows covering funding, final-model, validation and earlier-baseline metrics were verified against saved artifacts. No new experiment was run to produce this report.

Artifact index: `data/processed/xi_2027_63d_p50/`, `data/processed/rep_iran/`, `data/artifacts/xi_2027_63d_p50/` (funding tables, `null_model.json`,

`baseline_comparison.csv`, `final_test_results*.csv`, `tuning_*results.csv`), `data/artifacts/rep_iran/` (funding tables, `final_test_results*.csv`), `docs/funding_validation_findings.md` and `docs/tgnn_tuning_results.md`.

Appendix V

Hardware Specifications and Pin Diagram

This project is entirely software-based. Data were obtained from Polymarket's public trade interface and from Polygon transfer records, and all processing, training and evaluation were performed on a general-purpose computer, with final model runs on CPU. No sensor, microcontroller, embedded board or custom circuit was designed or used; therefore a pin diagram is not applicable. The computing environment is summarised in Table 3 of Chapter 3; the workstation processor, memory and operating system should be recorded there before final submission.